

# ASTR/PTYS 578: Assignment 4 Part A,B & C

Part A& B DUE : Feb 11th 2026 by **NOON**

FORMAT: Word doc

FILENAME: LastName\_Assignment4\_PartAB

## **Part A: Figure and Caption**

Construct a **figure** and **caption** that outlines a strategy for a solution to a problem that you will use HST/JWST/Theoretical Model to solve.

i.e. the figure and caption convey: Target + Strategy = Solution.

### **Figure Details:**

- The figure can be a diagram or sketch – you can design it by hand/power point etc. It does not have to be the final polished version that you will use in the proposal.
- You can modify a figure from a published paper – make sure to cite the paper.
- The figure can have multiple panels.
- Somewhere in the figure itself should be the words: “This proposal” to highlight the strategy being proposed.

### **Figure caption must follow the following format:**

- First sentence: bold faced takeaway that highlights the broader impact of the program. Effectively states the outcome of the program.
- Second(+) sentence(s): Description of what is plotted
- Following the description is a statement of the problem related to the target, which will be solved by the plot.
- Last sentence(s) is the Goal. Must include : “This program will” ... or “We propose to” AND the importance of the solution to the subfield. Underline the goal.

## **Part B: Talk Abstract about the Figure**

Create a talk abstract centered on your figure. Include this in the doc after the Part A.

- 1) Determine a talk title that is connected to “the problem” that is solved by your figure from Part A. Follow the “[Big Picture]: [Direct Problem]” format.
- 2) Construct a talk abstract as follows:
  - One sentence : one big picture Fact
  - One sentence : Importance of the Fact.
  - One sentence: Problem – this is **set up** by the first two sentences.
  - One Sentence: Goal: “In this talk I will discuss/demonstrate a \*strategy\* to solve \*the problem”. [ This maps on to “we propose” ]
    - If you don’t know the solution yet, that’s ok. you can write one idea down for now and we can build from there in class.
    - These Sites can help you figure out what you want to propose for:
      - JWST: <https://webb.nasa.gov/content/science/index.html>
      - HST: <https://www.stsci.edu/hst/about/key-science-themes>
  - One Sentence: Urgency/Importance to subfield. Why now? Why this facility/funding call?
  - One Sentence: Future Outlook/Broader Impact. End with a sentence about the importance of solving “the Problem” broadly to the field
  - Highlight each of the above components in your abstract in a **different color**.
  - Abstract must be no longer than **20 lines of 85 characters of text** (including spaces). In other words max 1700 characters including white spaces.

## **Part C: Presentation**

Create a presentation slide deck based on your figure and abstract, and present the slides in a short (7 min) talk to the class. The grade is based on the slide quality and presentation.

**FORMAT:** Oral presentation on schedule below & Upload a PDF of your presentation slides.

Format for Upload: "LastName\_Assignment4PartC.pdf"

### **Deadlines:**

- Upload your slides by March 5<sup>th</sup> at 5 PM on D2L.
- Presentation Schedule:
  - Feb 19<sup>th</sup>: Nick S., Carter M., Kayla S., Kylie H., Rafa B.
  - Feb 24<sup>nd</sup>: Maddy C., Ruby F., Devin H., Lily R., Tyler R.,
  - March 3<sup>rd</sup>: Aaron G, Alena C., Emily M, Maddy M., Yichen L
  - March 5<sup>th</sup>: Lipika C., Cole M., Neev S., Chen-Yu C., Sophie C.

**Duration:** 7 minutes + 3 min discussion

**Format/Instructions:** Turn your abstract into a 7 slide presentation (with relevant images):

**Slide 1)** Title Slide – spend 1 minute setting the stage!

Include: title, your name, one picture that is evocative

**Slide 2)** (1-2 slides max) Fact – this should correspond to the first sentences of your abstract. Set the background needed to understand the context for the problem

**Slide 3)** Importance of fact. (WHY does the topic MATTER – why should the audience care?)

**Slide 4)** The Target – what are you proposing to study/model?

**Slide 5)** The Problem – convey urgency in solving this \*now\*

**Slide 6)** Proposed Solution & Importance/Urgency (This will turn into the “we propose” e.g. Target + Proposed Data = Solution). This is where the Figure you created in Part A should be presented.

**Slide 7)** Broader Impacts/Future Outlooks: What future observatories/programs will benefit from your proposal?

- Each slide must have a heading that states the \*takeaway\* point for the slide.
- You must explain the axes for each figure.
- When asked questions, you must repeat the question back to the asker.

**Grading Criteria for the entirety of Assignment 4: (20% of final grade)**

- Part A: Follow the Structure Outlined Above for the Figure and Caption (/10)
- Part B: Follow the Structure Outlined Above for the Abstract (/10)
- Part C: (/20)
  - o Follow the Structure Outlined Above for the Presentation (/10)
  - o Quality of Presentation – clarity of slides & oral presentation (/10)Participation (asking **at least 3** questions to others)